

CSE2101 Software Engineering Fundamentals

Tutorial 4

Part A: Discussion

Topic (Lecture 4): Software Requirements Modelling

1. Give three reasons for requirements modelling.
2. Is it possible to begin coding immediately after a requirements model has been created?
3. Describe how the CRC cards can be used to identify classes in requirements analysis.
4. There are two different types of “states” that behavioural models can represent. What are they?
5. How does a sequence diagram differ from a state diagram? How are they similar?
6. What is the purpose of the interaction model for a Web/Mobile App?
7. What is the purpose of a configuration model?
8. How does the navigation model differ from the interaction model?

Part B: Exploration

Topic: Software Requirements Analysis

1. Refer to the following URL and discuss the usage and representations in Class Diagrams:
http://en.wikipedia.org/wiki/Class_diagram
2. Refer to the following URL and discuss the usage and representations in Entity-Relationship Diagrams:
http://en.wikipedia.org/wiki/Entity%E2%80%93relationship_model
3. Refer to the following URL and discuss the usage and representations in Class-Responsibility-Collaboration (CRC) cards:
https://en.wikipedia.org/wiki/Class-responsibility-collaboration_card
4. Identify a Web application that you often use. Discuss the following models for the Web application:
 - Content model
 - Interaction model
 - Functional model
 - Navigation model
 - Configuration model

5. Identify a mobile application that you often use. Similarly discuss the models for the mobile application.

Part C: Project

Start a Visual Paradigm project to save your project work in this tutorial. You can also use the project from previous tutorial.

1. For each use case assigned, review the description so that it clearly describes the interaction between the actor and the system, and the required processing. Ensure that at least the following items are identified:
 - the data to be entered by the actor,
 - the data or information to displayed/sent to the actor,
 - the data to be stored to the database/file,
 - the data to be retrieved from the database/file
2. Based on the data identified in the use cases, create the classes to represent the data. Draw a Class Diagram with all the classes in the system. Identify the attributes for each class based on the descriptions of data to be retrieved and stored in each class. Refer to the following webpage on how to draw class diagrams:

http://www.visual-paradigm.com/support/documents/vpuserguide/94/2576/7190_creatingclas.html
3. Identify the relationships between the objects, if any, and draw the relationships in the Class Diagram accordingly.